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| Surname       | Centre Number | Candidate Number |
| First name(s) |               | 0                |



GCSE  
3430UB0-1



FRIDAY, 16 JUNE 2023 – MORNING

SCIENCE (Double Award)  
Unit 2 – CHEMISTRY 1  
HIGHER TIER

1 hour 15 minutes

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **4** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

| For Examiner's use only |              |              |
|-------------------------|--------------|--------------|
| Question                | Maximum Mark | Mark Awarded |
| 1.                      | 8            |              |
| 2.                      | 7            |              |
| 3.                      | 4            |              |
| 4.                      | 6            |              |
| 5.                      | 8            |              |
| 6.                      | 9            |              |
| 7.                      | 9            |              |
| 8.                      | 9            |              |
| Total                   | 60           |              |

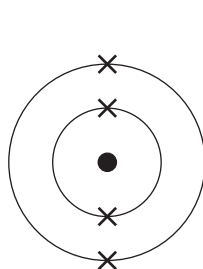


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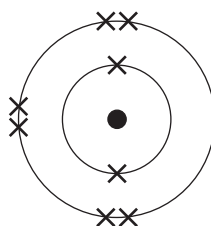
Answer **all** questions.

1. The diagrams below show the electronic structures of five elements, **A**, **B**, **C**, **D** and **E**.

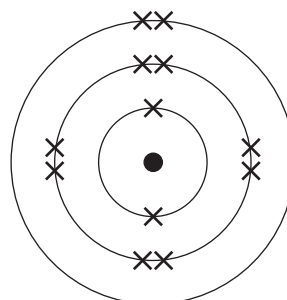
The letters are not the symbols of the elements.



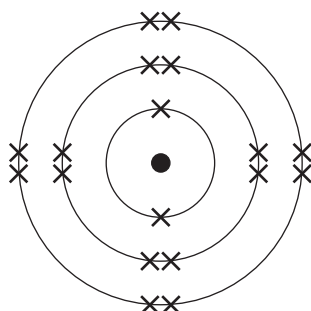
**A**



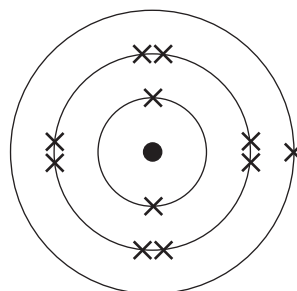
**B**



**C**



**D**



**E**

- (a) Give the **letters** of the elements found in Period 2 of the Periodic Table. Give a reason for your choice. [2]

Letters ..... and .....

Reason .....

.....



- (b) Give the **letter** of the element found in Group 0 of the Periodic Table. Give a reason for your choice. [2]

Letter .....

Reason .....

- (c) Explain how the electronic structure of element **E** can be used to determine its atomic number. [2]

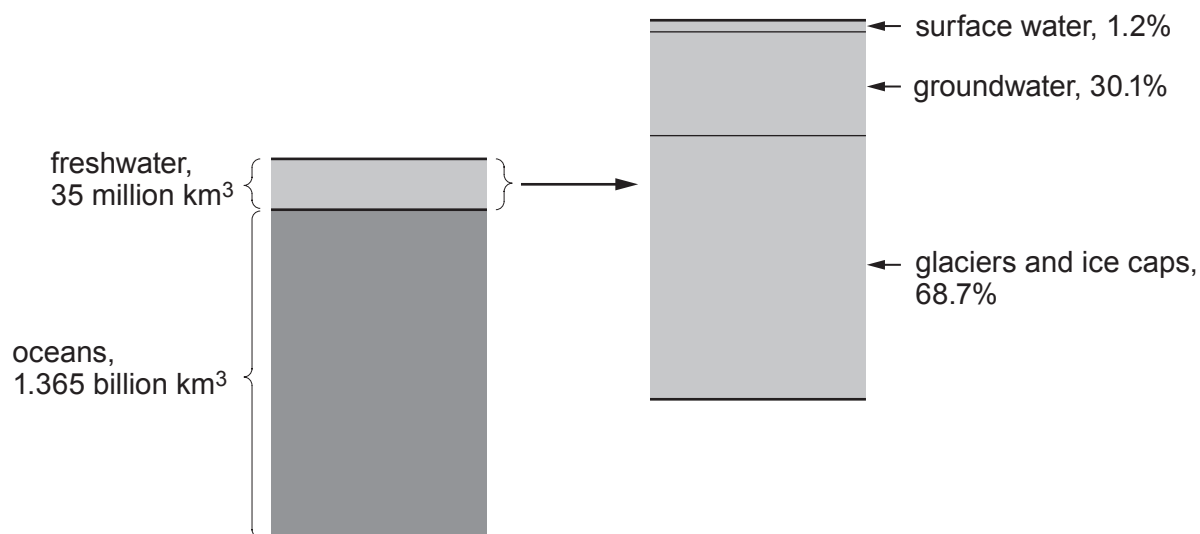
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- (d) One of the diagrams represents the element oxygen. Oxygen reacts with potassium to form potassium oxide.

Give the formula for potassium oxide and balance the equation for this reaction. [2]



2. The diagram shows the distribution of water on Earth.



- (a) Use the information in the diagram to calculate the volume of freshwater that is available as surface water. [2]

Volume = ..... million km<sup>3</sup>

- (b) As the Earth's population continues to increase, there is going to be a greater need to get freshwater from seawater. This is known as desalination.

Give a method that could be used to obtain freshwater from seawater. [1]

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(c) Calcium ions,  $\text{Ca}^{2+}$ , cause hardness in groundwater.

Describe an experiment to **compare** the hardness of two groundwater samples. [3]

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(d) Washing soda is used to soften hard water. This contains sodium carbonate.

Give the formula of sodium carbonate. [1]

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3. A food colouring contains a mixture of yellow and green dyes.
- The table shows the  $R_f$  values of these dyes using different solvents in separate chromatography experiments.

| Colour | $R_f$ value when the solvent is |      |         |
|--------|---------------------------------|------|---------|
|        | Water                           | Acid | Alcohol |
| yellow | 0.74                            | 0.35 | 0.00    |
| green  | 0.76                            | 0.81 | 0.46    |

- (a) Why would yellow and green dyes be difficult to separate in a chromatography experiment using **water** as the solvent? [1]

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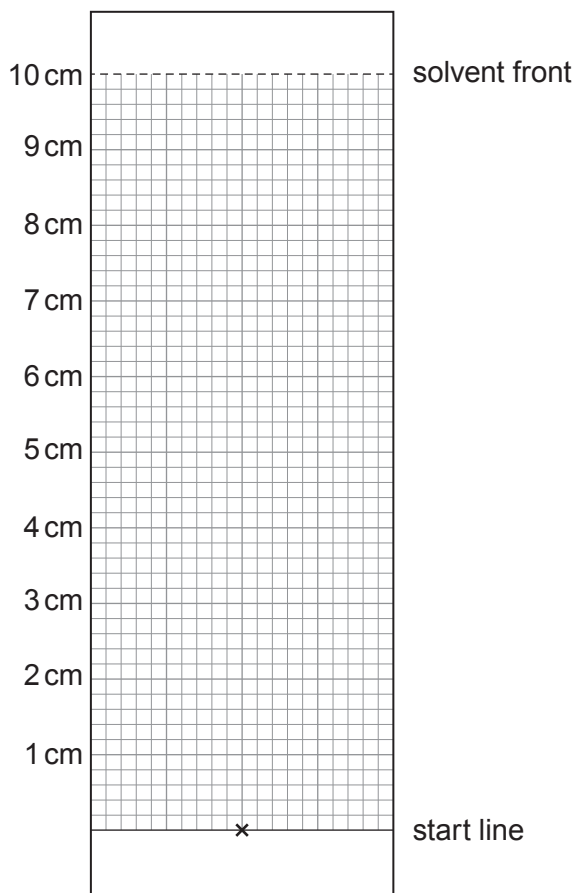
- (b) Describe what you would expect to see in the chromatogram obtained using **alcohol** as the solvent. [1]

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- (c) Complete the chromatogram to show the results you would expect using **acid** as the solvent. [2]







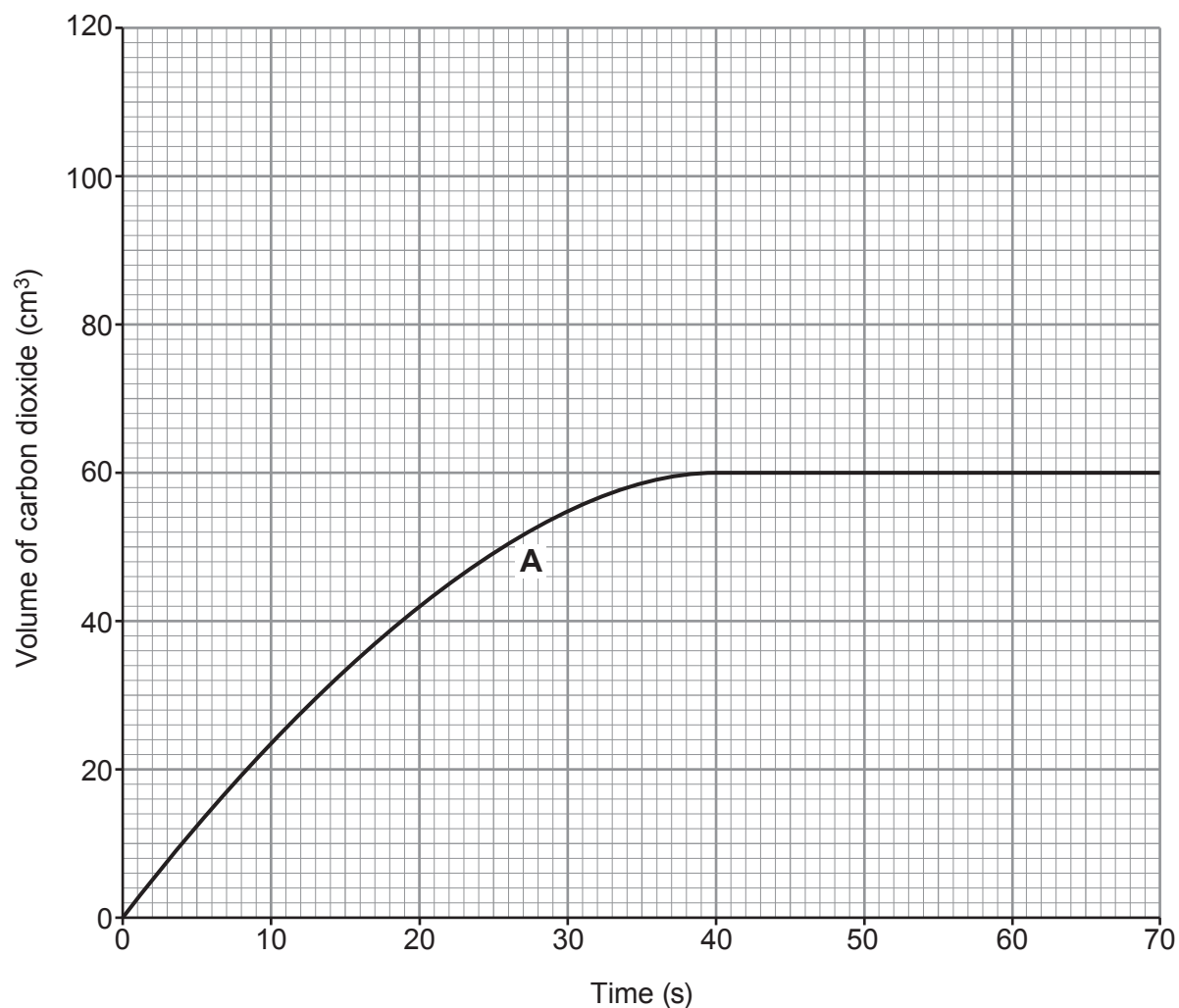
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ON THIS PAGE**



5. Marble chips (calcium carbonate) react with dilute hydrochloric acid to produce carbon dioxide gas.

Graph **A** shows the volume of carbon dioxide produced during the reaction between 0.25 g of marble chips and **excess** dilute hydrochloric acid at 30 °C.



- (a) State why no more marble chips remain when the reaction stops. [1]

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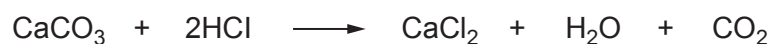
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- (b) The original experiment was repeated at 60 °C.

**On the grid** draw the curve you would expect to see for the production of carbon dioxide at this temperature. [1]



- (c) The equation for the reaction between calcium carbonate and dilute hydrochloric acid is as follows.



Calculate the mass of calcium chloride produced when 7.8 g of calcium carbonate is reacted with excess acid. [2]

$$M_r(\text{CaCO}_3) = 100$$

$$M_r(\text{CaCl}_2) = 111$$

Mass = ..... g

- (d) When the reaction was repeated with a different mass of calcium carbonate, it was found that 14.3 g of calcium chloride formed. This represents a yield of 53.7%.

Calculate the maximum mass of calcium chloride that could have been formed. [2]

Maximum mass = ..... g

- (e) Catalysts can be added to some chemical reactions in order to increase their rate.

Explain how a catalyst increases the rate of a chemical reaction. [2]

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6. Fluoride ions are added to drinking water in some areas of the UK.

The table below shows the effect of different levels of fluoride in drinking water on the number of decayed, missing and filled teeth (DMFT) and the percentage of people suffering from fluorosis.

| Concentration of fluoride<br>(mg/dm <sup>3</sup> ) | Mean DMFT | Percentage of people<br>affected by fluorosis (%) |
|----------------------------------------------------|-----------|---------------------------------------------------|
| 0.3                                                | 7.0       | 4                                                 |
| 0.6                                                | 4.5       | 6                                                 |
| 0.9                                                | 3.0       | 15                                                |
| 1.2                                                | 2.5       | 35                                                |
| 1.5                                                | 2.3       | 40                                                |
| 1.8                                                | 2.0       | 45                                                |
| 2.1                                                | 2.1       | 60                                                |
| 2.4                                                | 2.1       | 68                                                |
| 2.7                                                | 2.1       | 75                                                |

- (a) The recommended concentration of fluoride ions to be added to drinking water in the UK is 1.0 mg/dm<sup>3</sup>.

**Use the information in the table** to explain why this is the recommended level in terms of DMFT and fluorosis. [4]

DMFT

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Fluorosis

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- (b) Chlorine is added to all drinking water in the UK, whilst fluoride is only added in some areas.

Explain why some people are opposed to adding fluoride but no-one is opposed to adding chlorine to drinking water. [2]

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- (c) Fluorine reacts with titanium at very high temperatures, forming titanium fluoride.

In an experiment 3.8 g of fluorine reacted with 2.4 g of titanium.

Calculate the simplest formula for titanium fluoride. You **must** show your working. [3]

$$A_r(\text{Ti}) = 48$$

$$A_r(\text{F}) = 19$$

Simplest formula .....

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7. A great deal of scientific research into global warming has happened over the last 30 years.

More recently, two new contrasting theories linked to global warming have been widely reported. These are outlined below.

**Solar activity**

A small number of scientists believe that changes in solar activity (changes in the brightness and warmth of the Sun) is causing global warming. Solar activity occurs on the surface of the Sun when nuclear reactions cause flares to be released, high winds to occur and a release of high energy particles from the Sun.

**Global dimming**

Some scientists believe that global warming is increasing the Earth’s mean temperature, but that global dimming is preventing it from increasing even more. Global dimming is the reduction in the amount of radiation reaching the Earth’s surface from the Sun. It is thought to be caused by an increase of aerosol particles in the Earth’s atmosphere caused by pollution, dust and smog, as well as volcanic eruptions. These particles absorb solar energy and reflect sunlight back into space.

**Figure 1** shows the mean atmospheric temperature, carbon dioxide concentration in the atmosphere and solar activity between 1930 and 2010.

**Figure 1**

| Year | Mean atmospheric temperature (°C) | CO <sub>2</sub> concentration (ppm) | Solar activity (arbitrary units) |
|------|-----------------------------------|-------------------------------------|----------------------------------|
| 1930 | 13.6                              | 309                                 | 3.9                              |
| 1940 | 13.6                              | 311                                 | 4.0                              |
| 1950 | 13.8                              | 316                                 | 4.2                              |
| 1960 | 13.9                              | 320                                 | 4.2                              |
| 1970 | 14.0                              | 331                                 | 4.0                              |
| 1980 | 14.2                              | 342                                 | 3.9                              |
| 1990 | 14.3                              | 358                                 | 3.7                              |
| 2000 | 14.4                              | 370                                 | 3.7                              |
| 2010 | 14.6                              | 383                                 | 3.6                              |

ppm = parts per million



**Figure 2** shows the mean atmospheric temperature within a 20-mile radius of the Mount Pinatubo volcano in the Philippines between 1990 and 2000. Mount Pinatubo erupted in 1991.

**Figure 2**

| Year | Mean atmospheric temperature (°C) |
|------|-----------------------------------|
| 1990 | 15.1                              |
| 1991 | 14.7                              |
| 1992 | 14.8                              |
| 1993 | 14.7                              |
| 1994 | 14.7                              |
| 1995 | 14.6                              |
| 1996 | 14.7                              |
| 1997 | 14.8                              |
| 1998 | 14.7                              |
| 1999 | 14.8                              |
| 2000 | 14.9                              |

- (a) Tick (✓) the box that **best** describes the change in carbon dioxide concentration in the atmosphere between 1930 and 2010. [1]

Carbon dioxide concentration increased by approximately 10 ppm every 10 years

☐

Carbon dioxide concentration increased more between 1970 and 2010 than it did between 1930 and 1960

☐

Carbon dioxide concentration increased more between 1930 and 1960 than it did between 1970 and 2010

☐

There is no trend to the change in carbon dioxide concentration between 1930 and 2010

☐


- (b) A newspaper report states that global warming is caused by increases in atmospheric carbon dioxide levels **and** by increases in solar activity.

Explain whether the data in **Figure 1** supports this claim. [2]

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- (c) **Figure 2** shows that the mean atmospheric temperature near Mount Pinatubo decreased after the volcanic eruption of 1991. Suggest why scientists would **not** use these findings to explain how volcanic activity affects **global** temperatures. [1]

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- (d) State **one** industrial method that is being developed to reduce the concentration of carbon dioxide in the atmosphere. [1]

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- (e) The Earth's original atmosphere contained about 95% carbon dioxide. Today, the atmosphere contains 0.04% carbon dioxide. Explain why the level of carbon dioxide decreased over geological time. [3]

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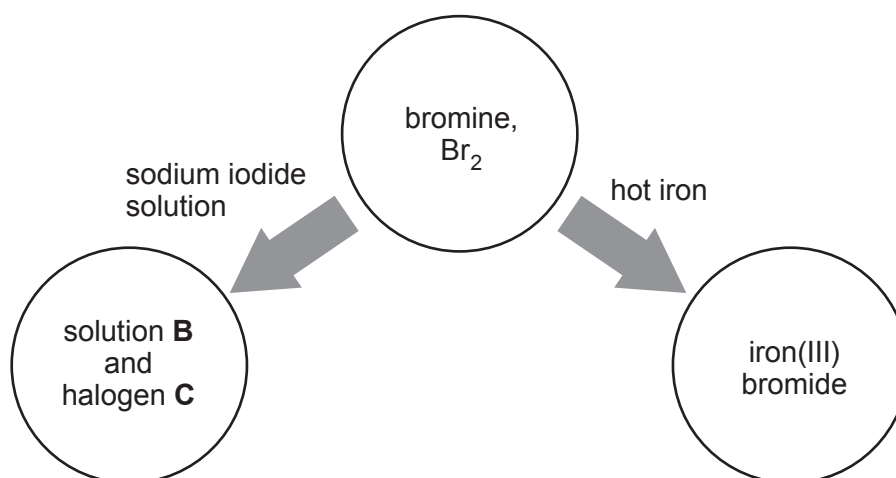
- (f) When carbon dioxide turns limewater milky, calcium hydroxide is produced.  
Give the formula of calcium hydroxide. [1]

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8. The diagram shows reactions of bromine,  $\text{Br}_2$ .



- (a) (i) Give the names of solution **B** and halogen **C**. [2]

solution **B** .....

halogen **C** .....

- (ii) Explain why bromine reacts with sodium iodide solution. [2]

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- (b) Write the symbol equation for the reaction between bromine,  $\text{Br}_2$ , and iron to form iron(III) bromide. [2]

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- (c) Silver nitrate solution was added to a solution of sodium iodide. Silver iodide was formed.

- (i) State what you would expect to see. [1]

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- (ii) Give the **ionic** equation for the formation of silver iodide. [2]

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**END OF PAPER**



[illegible]

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### FORMULAE FOR SOME COMMON IONS

| POSITIVE IONS |                  | NEGATIVE IONS |                    |
|---------------|------------------|---------------|--------------------|
| Name          | Formula          | Name          | Formula            |
| aluminium     | $\text{Al}^{3+}$ | bromide       | $\text{Br}^-$      |
| ammonium      | $\text{NH}_4^+$  | carbonate     | $\text{CO}_3^{2-}$ |
| barium        | $\text{Ba}^{2+}$ | chloride      | $\text{Cl}^-$      |
| calcium       | $\text{Ca}^{2+}$ | fluoride      | $\text{F}^-$       |
| copper(II)    | $\text{Cu}^{2+}$ | hydroxide     | $\text{OH}^-$      |
| hydrogen      | $\text{H}^+$     | iodide        | $\text{I}^-$       |
| iron(II)      | $\text{Fe}^{2+}$ | nitrate       | $\text{NO}_3^-$    |
| iron(III)     | $\text{Fe}^{3+}$ | oxide         | $\text{O}^{2-}$    |
| lithium       | $\text{Li}^+$    | sulfate       | $\text{SO}_4^{2-}$ |
| magnesium     | $\text{Mg}^{2+}$ |               |                    |
| nickel        | $\text{Ni}^{2+}$ |               |                    |
| potassium     | $\text{K}^+$     |               |                    |
| silver        | $\text{Ag}^+$    |               |                    |
| sodium        | $\text{Na}^+$    |               |                    |
| zinc          | $\text{Zn}^{2+}$ |               |                    |



THE PERIODIC TABLE

Group

1 2 3 4 5 6 7 0

|                                                             |                             |                              |                             |                             |                              |                              |                              |                            |                              |                            |                            |                             |                             |                             |                              |                             |                           |                                                            |                         |
|-------------------------------------------------------------|-----------------------------|------------------------------|-----------------------------|-----------------------------|------------------------------|------------------------------|------------------------------|----------------------------|------------------------------|----------------------------|----------------------------|-----------------------------|-----------------------------|-----------------------------|------------------------------|-----------------------------|---------------------------|------------------------------------------------------------|-------------------------|
| <div><div><div>1<br/>H<br/>Hydrogen<br/>1</div></div></div> |                             |                              |                             |                             |                              |                              |                              |                            |                              |                            |                            |                             |                             |                             |                              |                             |                           | <div><div><div>4<br/>He<br/>Helium<br/>2</div></div></div> |                         |
| 7<br>Li<br>Lithium<br>3                                     | 9<br>Be<br>Beryllium<br>4   |                              |                             |                             |                              |                              |                              |                            |                              |                            |                            |                             |                             |                             |                              |                             |                           | 19<br>F<br>Fluorine<br>9                                   | 20<br>Ne<br>Neon<br>10  |
| 23<br>Na<br>Sodium<br>11                                    | 24<br>Mg<br>Magnesium<br>12 |                              |                             |                             |                              |                              |                              |                            |                              |                            |                            |                             |                             |                             |                              |                             |                           | 35.5<br>Cl<br>Chlorine<br>17                               | 40<br>Ar<br>Argon<br>18 |
| 39<br>K<br>Potassium<br>19                                  | 40<br>Ca<br>Calcium<br>20   | 45<br>Sc<br>Scandium<br>21   | 48<br>Ti<br>Titanium<br>22  | 51<br>V<br>Vanadium<br>23   | 52<br>Cr<br>Chromium<br>24   | 55<br>Mn<br>Manganese<br>25  | 56<br>Fe<br>Iron<br>26       | 59<br>Co<br>Cobalt<br>27   | 59<br>Ni<br>Nickel<br>28     | 63.5<br>Cu<br>Copper<br>29 | 65<br>Zn<br>Zinc<br>30     | 70<br>Ga<br>Gallium<br>31   | 73<br>Ge<br>Germanium<br>32 | 75<br>As<br>Arsenic<br>33   | 79<br>Se<br>Selenium<br>34   | 80<br>Br<br>Bromine<br>35   | 84<br>Kr<br>Krypton<br>36 |                                                            |                         |
| 86<br>Rb<br>Rubidium<br>37                                  | 88<br>Sr<br>Strontium<br>38 | 89<br>Y<br>Yttrium<br>39     | 91<br>Zr<br>Zirconium<br>40 | 93<br>Nb<br>Niobium<br>41   | 96<br>Mo<br>Molybdenum<br>42 | 99<br>Tc<br>Technetium<br>43 | 101<br>Ru<br>Ruthenium<br>44 | 103<br>Rh<br>Rhodium<br>45 | 106<br>Pd<br>Palladium<br>46 | 108<br>Ag<br>Silver<br>47  | 112<br>Cd<br>Cadmium<br>48 | 115<br>In<br>Indium<br>49   | 119<br>Sn<br>Tin<br>50      | 122<br>Sb<br>Antimony<br>51 | 128<br>Te<br>Tellurium<br>52 | 127<br>I<br>Iodine<br>53    | 131<br>Xe<br>Xenon<br>54  |                                                            |                         |
| 133<br>Cs<br>Caesium<br>55                                  | 137<br>Ba<br>Barium<br>56   | 139<br>La<br>Lanthanum<br>57 | 179<br>Hf<br>Hafnium<br>72  | 181<br>Ta<br>Tantalum<br>73 | 184<br>W<br>Tungsten<br>74   | 186<br>Re<br>Rhenium<br>75   | 190<br>Os<br>Osmium<br>76    | 192<br>Ir<br>Iridium<br>77 | 195<br>Pt<br>Platinum<br>78  | 197<br>Au<br>Gold<br>79    | 201<br>Hg<br>Mercury<br>80 | 204<br>Tl<br>Thallium<br>81 | 207<br>Pb<br>Lead<br>82     | 209<br>Bi<br>Bismuth<br>83  | 210<br>Po<br>Polonium<br>84  | 210<br>At<br>Astatine<br>85 | 222<br>Rn<br>Radon<br>86  |                                                            |                         |
| 223<br>Fr<br>Francium<br>87                                 | 226<br>Ra<br>Radium<br>88   | 227<br>Ac<br>Actinium<br>89  |                             |                             |                              |                              |                              |                            |                              |                            |                            |                             |                             |                             |                              |                             |                           |                                                            |                         |

Key

A<sub>r</sub>

Symbol

Name

Z

relative atomic mass

atomic number

